

I'm a Cornell Computer Science PhD student co-advised by David Shmoys and Andrea Lodi. My research focuses on developing optimization and AI methods for real-world systems, including problems in transportation and resource management. Previously, I worked with Carla Gomes to develop methods to monitor aquaculture development from satellite imagery, with Claire Kremen to scale an ecological connectivity analysis to the global level, and with Geoffrey Schiebinger on optimization methods for single-cell genomics.

EDUCATION

- 2021 - Present **Ph.D. Computer Science**, Cornell University
Research areas: Operations research, optimization, deep learning
- 2020 **B.Sc. Honours Computer Science, Mathematics Minor**,
University of British Columbia

SELECTED PROJECTS

- 05/2025 - Present **Optimization and Simulation to Improve NYC Buses**
Through a Cornell Siegel PiTech Impact fellowship, I created a simulation framework using MTA open data to understand what factors drive poor performance and compare interventions. I am now partnering with the MTA to use the simulation internally and design interventions.
- 05/2024 - Present **Optimization for Fisheries Planning, Cornell University**
Working with fisheries experts to model and design interventions to reduce community-level fishery income variability in Alaska using optimization and mathematical programming methods.
Greenstreet, L., Q. Shi, M. Grimson, F.W. Simon, S.A. Sethi, D. Shmoys, A. Lodi. Reducing Income Variability in Natural Resource Portfolios via Integer Programming. *CPAIOR*, 2025. https://doi.org/10.1007/978-3-031-95976-9_2
- 09/2022 - 05/2024 **Deep Learning for Aquaculture Mapping in the Amazon, Cornell University**
Developed deep learning models to map aquaculture development from satellite imagery. Models were used by ecologists and the Brazilian government to understand environmental impacts and development trade-offs.
Greenstreet, L., J. Fan, F. Siqueira Pacheco, Y. Bai, M. Eichemberger Ummus, et al. Detecting Aquaculture with Deep Learning in a Low-Data Setting. *SigKDD 2023 Fragile Earth Workshop*.
F.S. Pacheco, S.A. Heilpern, C. DiLeo, ..., L. Greenstreet, et al. Towards sustainable aquaculture in the Amazon. *Nature Sustainability*, 2025. <https://doi.org/10.1038/s41893-024-01500-w>
- 05/2020 - 09/2021 **Trajectory Inference for Single-Cell Data, University of British Columbia**
Developed mathematical trajectory inference methods to understand spatial and temporal patterns in genomic single-cell sequencing data. Worked closely with medical researchers and biologists to demonstrate the methods in multiple biological systems.

05/2020 - 09/2021 **Trajectory Inference for Genomic Single-Cell Data, University of British Columbia cont.**

Greenstreet, L., A. Afanassiev, Y. Kijima, M. Heitz, S. Ichiguro, et al. *DNA-GPS: A Theoretical Framework for Optics-Free Spatial Genomics and Synthesis of Current Methods*. *Cell Systems*, 2023. <https://doi.org/10.1016/j.cels.2023.08.005>

Zhang, S., A. Afanassiev, **L. Greenstreet**, T. Matsumoto, G. Schiebinger. *Optimal transport analysis reveals trajectories in steady-state systems*. *PLOS Computational Biology*, 2021. <https://doi.org/10.1371/journal.pcbi.1009466>

T.M. Nolan, N. Vukasinovic, C.W. Hsu, J. Zhang, ..., **L. Greenstreet**, et al. *Brassinosteroid gene regulatory networks at cellular resolution in the Arabidopsis root*. *Science*, 2023. <https://doi.org/10.1126/science.adf4721>

05/2019 - 08/2019 **Electric Vehicle Infrastructure Planning, Data Science for Social Good Fellowship, University of British Columbia**

Worked with planners from the City of Surrey, British Columbia to inform their electric vehicle strategy by integrating six datasets, developing an app to visualize and analyze data, and developing a charger placement model. Followed up with the city from 2020-2022 to update and integrate the model with city systems.

Greenstreet, L, E.Y. Lai, K. Lin, G.A. Rodriguez-Arelis, R. Ng. Developing a Smart Electric Vehicle Strategy: From Data to Decisions. *ICDE Workshop on Data-Driven Smart Cities*.

Greenstreet, L, and E. Lai. *Developing a Data-Driven Electric Vehicle Strategy in Surrey, BC. SigKDD 2020 Social Impact Session*.

AWARDS

2025	Siegel PiTech PhD Impact Fellowship , Department of Computer Science, Cornell University
2022-2026	Graduate Teaching Award x5 , Department of Computer Science, Cornell University
2020	Undergraduate Summer Research Award , Natural Sciences and Engineering Research Council of Canada (NSERC)
2019	Data Science for Social Good Fellowship , UBC Data Science Institute

TECHNICAL SKILLS

4+ Years Experience: Python, Git, Linux

1-3 Years Experience: R, Julia, Matlab, Java, SQL

Libraries/Tools: Pytorch, QGIS, Postgres, Gurobi

Experience: deep learning: CNNs, GNNs, VAEs, transformers, contrastive learning; manifold learning; mathematical programming; SQL

COMMUNITY INVOLVEMENT

2024	Organizer , AI for Science Seminar
2023	Organizer , NeurIPS Computational Sustainability Workshop
2023	Mentor , BURE Undergraduate Research Program, Cornell
2023-2024	Assistant Organizer , AI for Science Program, Cornell
2020-2021	Mentor , Data Science for Social Good Program, UBC Data Science Institute

ADDITIONAL PUBLICATIONS

- 2025 Li, H., P. Cote, M. Kuoch, J. Ezike, ..., **L. Greenstreet**, et al. *The dynamics of hematopoiesis over the human lifespan*. Nature Methods, 2025. <https://doi.org/10.1038/s41592-024-02495-0>
- 2025 Massri, A.J., A. Berrio, A. Afanassiev, **L. Greenstreet**, et al. *Single-Cell Transcriptomics Reveals Evolutionary Reconfiguration of Embryonic Cell Fate Specification in the Sea Urchin *Heliocidaris erythrogramma**. Genome Biology and Evolution, 2025. <https://doi.org/10.1093/gbe/evae258>
- 2023 Mirka, R., **L. Greenstreet**, M. Grimson, C.P. Gomes. *A New Approach to Finding $2 \times n$ Partially Spatially Balanced Latin Rectangles*, CP, 2023.
- 2022 Brennan, A., R. Naidoo, **L. Greenstreet**, Z. Mehrabi, N. Ramankutty, C. Kremen. *Functional Connectivity of the Worlds Protected Areas*. Science, 2022. <https://doi.org/10.1126/science.abl8974>
- 2022 **Greenstreet, L.**, N.J.A. Harvey, V. Sanches Portella. *Efficient and Optimal Fixed-Time Regret with Two Experts*. ALT, 2022. <https://doi.org/10.48550/arXiv.2203.07577>
- 2021 Li, H., J. Ezike, A. Afanassiev, **L. Greenstreet**, et al. *Single Cell Analysis Elucidates the Maturation of Human Stem and Progenitor Cell Function from Fetal through Adult Hematopoiesis*. Blood, 2021. <https://doi.org/10.1182/blood-2021-151090>
- 2021 Shahan R., C. Hsu, T.M. Nolan, B.J. Cole, I.W. Taylor, **L. Greenstreet**, et al. *A single cell Arabidopsis root atlas reveals developmental trajectories in wild type and cell identity mutants*. Developmental Cell, 2021. <https://doi.org/10.1016/j.devcel.2022.01.008>
- 2020 Massri, A.J., **L. Greenstreet**, A. Afanassiev, A. Berrio Escobar, G.M. Wray, G. Schiebinger, D.R. McClay. *Developmental Single-cell transcriptomics in the *Lytechinus variegatus* Sea Urchin Embryo*. Development, 2020. <https://doi.org/10.1242/dev.198614>